

STD – X
Subject : Science



CHAPTER 1
**Chemical Reactions
and Equations**





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Chemical Reactions and Equations

Part – 5

Types of Chemical Reactions

C) Displacement Reaction

A chemical reaction in which a more reactive element displaces a less reactive element from its compound (or salt solution) is known as displacement reaction.



C) Displacement Reaction

A chemical reaction in which a more reactive element displaces a less reactive element from its compound (or salt solution) is known as displacement reaction.

Element	Symbol	
Potassium	K	
Sodium	Na	
Calcium	Ca	
Magnesium	Mg	
Aluminium	Al	
Zinc	Zn	
Iron	Fe	
Tin	Sn	
Lead	Pb	
Hydrogen	H	
Copper	Cu	
Mercury	Hg	
Silver	Ag	
Gold	Au	

More reactive →

→ Standard Reference

Less reactive →

Most Reactive

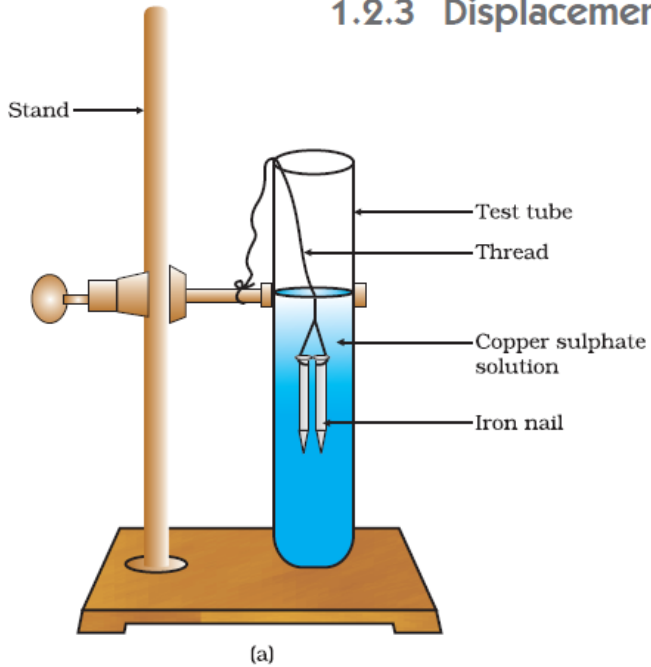
Decreasing Chemical Reactivity

Least Reactive

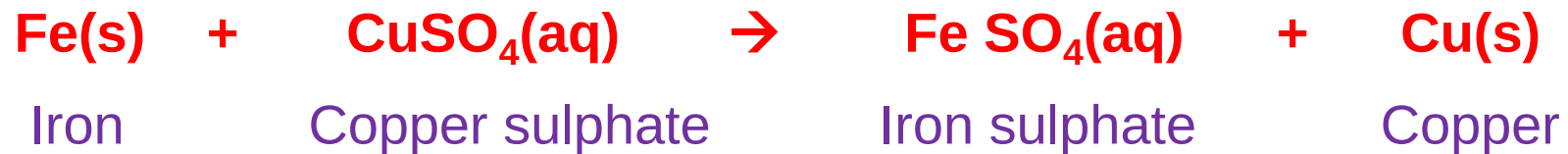
C) Displacement Reaction

Activity – 1.9 : Iron nail and Copper Sulphate

1.2.3 Displacement

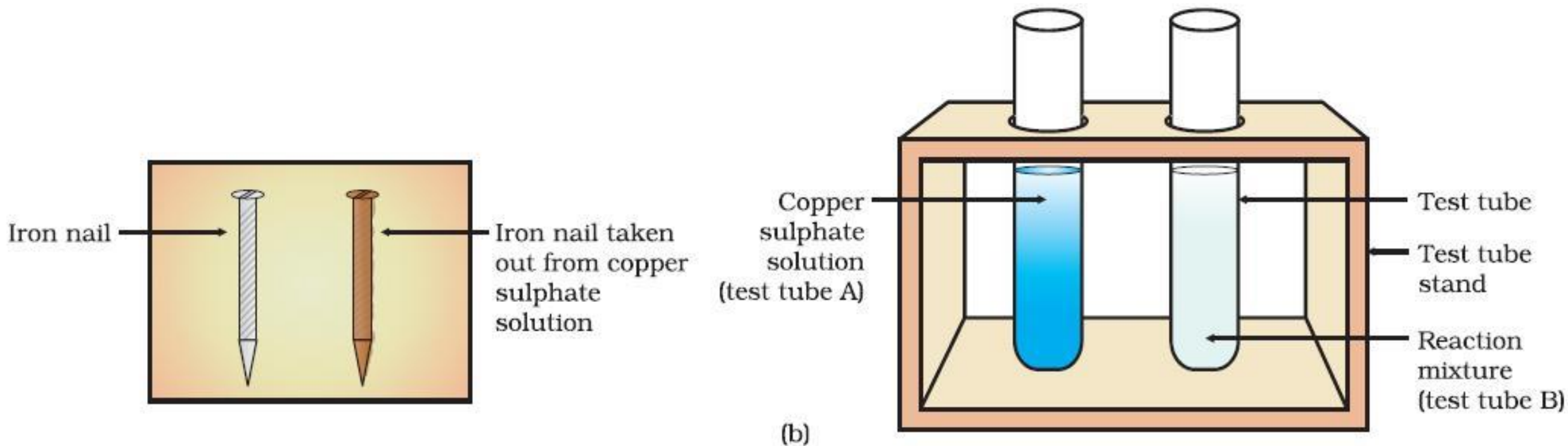


• Chemical Equation :

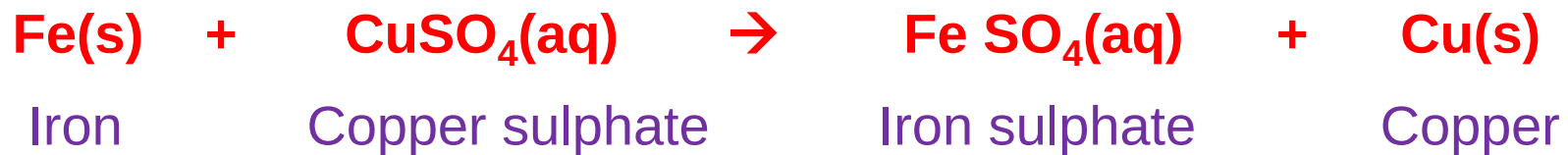


C) Displacement Reaction

Activity – 1.9 : Iron nail and Copper Sulphate



• Chemical Equation :

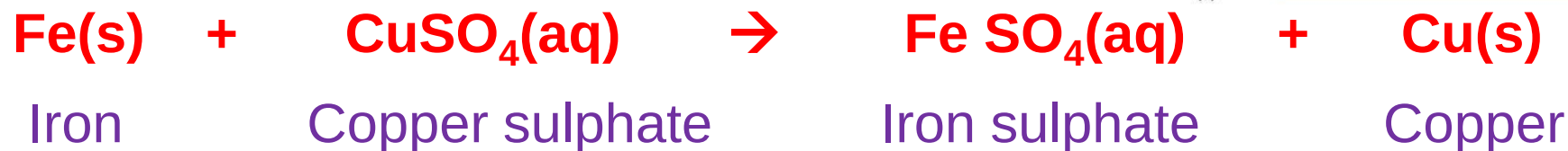


C) Displacement Reaction

- Observations :

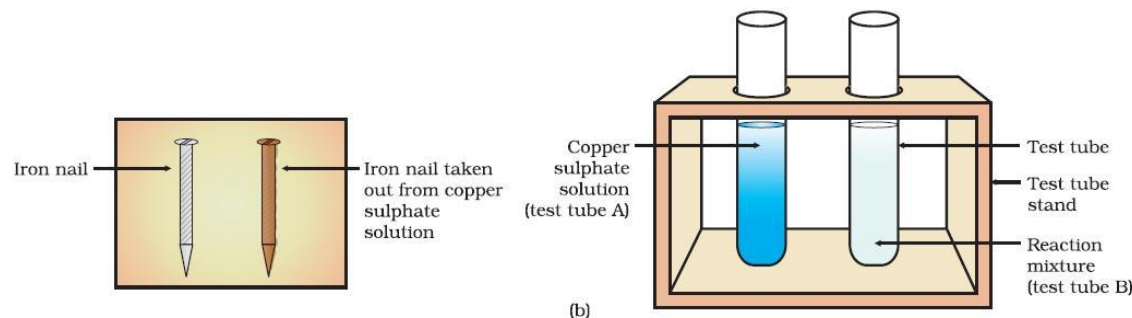
- 1) Iron nail becomes brownish in colour due to deposition of copper on it.
- 2) The blue colour of the copper sulphate solutions fades to light green.

- Chemical Equation :



- Conclusion / Inference:

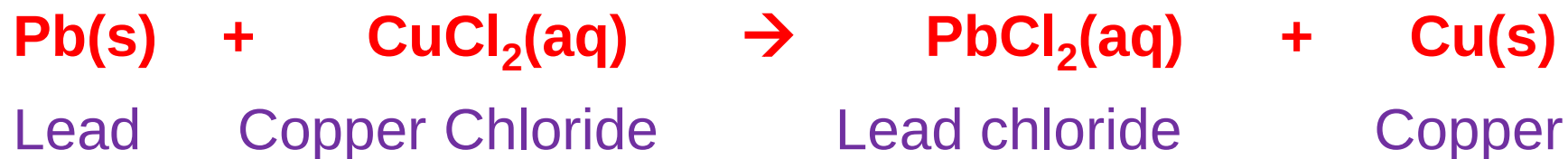
- 1) Iron is more reactive metal than copper.
- 2) Iron displaces copper from the copper sulphate solution.
- 3) Hence, This is a displacement reaction.



C) Displacement Reaction

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Potassium	K
Sodium	Na
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Zinc	Zn
Iron	Fe
Tin	Sn
Lead	Pb
Hydrogen	H
Copper	Cu
Mercury	Hg
Silver	Ag
Gold	Au

- Other Examples of displacement reactions :



- Zinc and lead are more reactive elements than copper.
- They displace copper from its compounds.

TO BE CONTINUED

